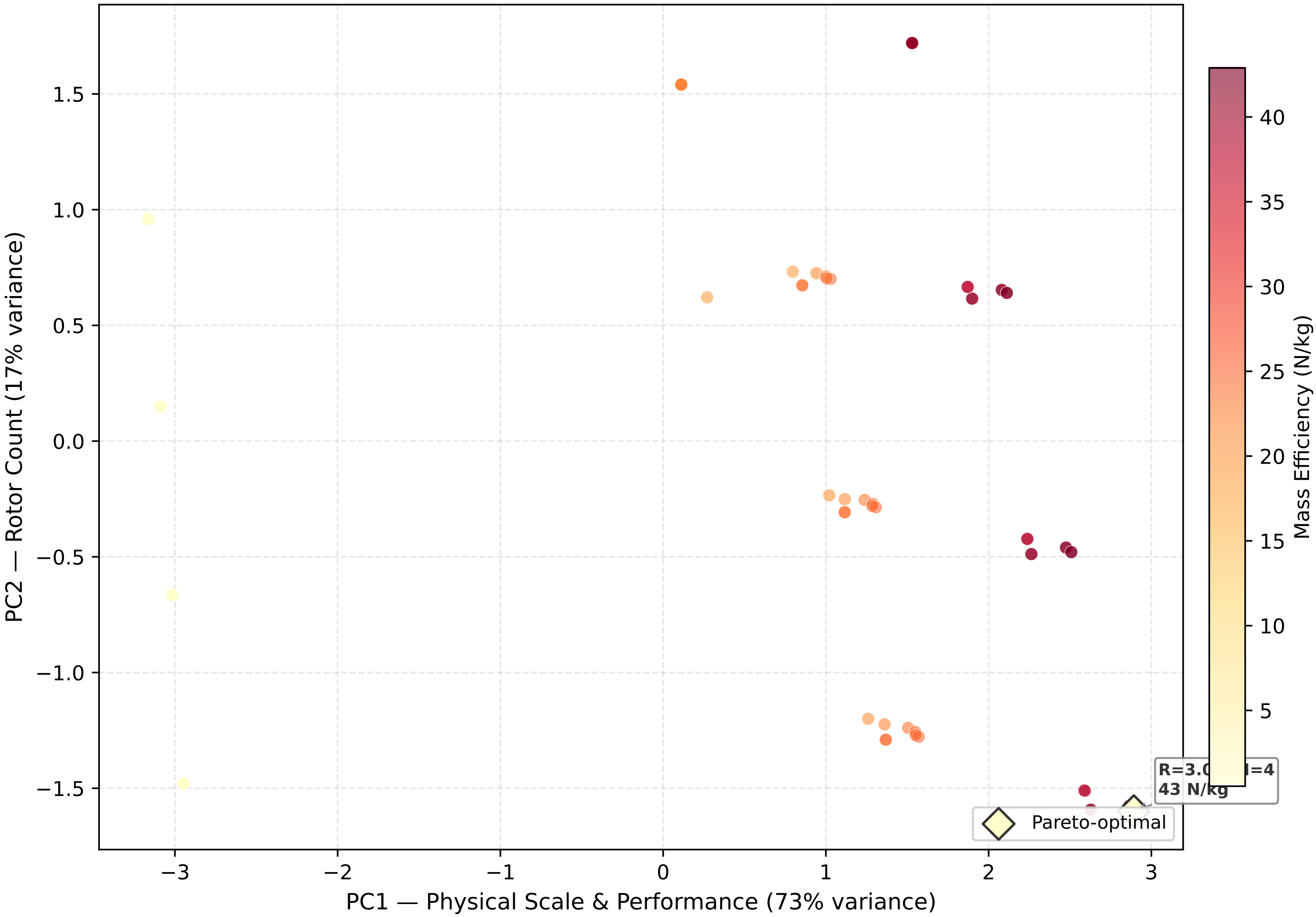
Mass Efficiency in PC Space: Where's the Best Bang-for-Buck?



IMPLICATIONS: Mass efficiency (N/kg) maps cleanly onto PC space — higher PC1 (larger radius, faster tips) yields higher N/kg, while PC2 (rotor count) distributes efficiency within each radial band. The Pareto frontier (diamond markers) traces the efficient frontier: designs that maximise N/kg for their position in design space. Key finding: R=3.0m dominates the Pareto frontier. For the pitch deck: 'every design on this curve gives best bang-for-buck at its scale.' The frontier is smooth and convex — no awkward gaps, no diminishing returns cliff. Data: 96 configurations from BEM v2.0 sweep, 1 Pareto-optimal.